Accounting Analysis

Contents & Quick Review

#	Topic	Correct Answer
1	Gross Profit Ratio	**$120,000 gross profit; 40%**
2	Current Ratio	**2:1**
3	Quick Ratio	**$100,000 quick assets; 2:1**
4	Net Profit Ratio	**($40,000 ÷ $200,000) × 100 = 20%**
5	Working Capital	**$60,000**
6	Inventory Turnover Ratio	**6 times**
7	Debt to Equity Ratio	**2:1**
8	ROCE	**20%**
9	EPS	**$4.00 per share**
10	Accounts Receivable Turnover Ratio	**5 times**

Error patterns found

- **Reversed formula / division —** Items 4, 6, 7, 8, 9, 10
- **Wrong arithmetic —** Items 1 and 2
- **Wrong operation / sign —** Item 5
- **Wrong treatment of inventory —** Item 3

01 Gross Profit Ratio

Question data

- Net Sales = \$300,000
- Cost of Goods Sold = \$180,000

Supplied solution

Gross Profit: 300,000 − 180,000 = 100,000; Gross Profit Ratio = (100,000 / 300,000) × 100 = 25%

✕ MISTAKE IDENTIFIED

Gross profit was calculated incorrectly. \$300,000 − \$180,000 = \$120,000, not \$100,000.

✓ CORRECT FORMULA

Gross Profit = Net Sales − COGS; Gross Profit Ratio = (Gross Profit ÷ Net Sales) × 100

Reasoning

First calculate the profit correctly. Then divide gross profit by net sales because the ratio measures the profit earned from sales.

✓ CORRECT RESULT

\$120,000 gross profit; 40%

02 Current Ratio

Question data

- Current Assets = $80,000
- Current Liabilities = $40,000

Supplied solution

80,000 / 40,000 = 1:1

MISTAKE IDENTIFIED

The division is correct, but the final ratio is not. $80,000 ÷ $40,000 = 2, so the ratio is 2:1.

CORRECT FORMULA

Current Ratio = Current Assets ÷ Current Liabilities

Reasoning

Compare the resources available to meet short-term obligations with the obligations themselves.

CORRECT RESULT

2:1

03 Quick Ratio

Question data

- Current Assets = $120,000
- Inventory = $20,000
- Current Liabilities = $50,000

Supplied solution

Quick Assets: 120,000 + 20,000 = 140,000; Quick Ratio = 140,000 / 50,000 = 2:1

× **MISTAKE IDENTIFIED**

Inventory should be excluded from quick assets, not added. Quick assets = $120,000 − $20,000 = $100,000.

✓ **CORRECT FORMULA**

Quick Assets = Current Assets − Inventory; Quick Ratio = Quick Assets ÷ Current Liabilities

Reasoning

The quick ratio focuses on assets that can be used more readily to meet current liabilities, so inventory is removed.

✓ **CORRECT RESULT**

$100,000 quick assets; 2:1

04 Net Profit Ratio

Question data

- Net Profit = $40,000
- Net Sales = $200,000

Supplied solution

200,000 / 40,000 × 100 = 50%

✕ **MISTAKE IDENTIFIED**

The numerator and denominator were reversed. Net profit must be divided by net sales.

✓ **CORRECT FORMULA**

Net Profit Ratio = (Net Profit ÷ Net Sales) × 100

Reasoning

Profitability is expressed as the portion of sales retained as net profit.

✓ **CORRECT RESULT**

($40,000 ÷ $200,000) × 100 = 20%

05 Working Capital

Question data

- Current Assets = $90,000
- Current Liabilities = $30,000

Supplied solution

90,000 + 30,000 = 120,000; Answer: $120,000

MISTAKE IDENTIFIED

Working capital is the excess of current assets over current liabilities, so subtraction is required.

CORRECT FORMULA

Working Capital = Current Assets − Current Liabilities

Reasoning

It shows the net short-term resources left after current obligations are covered.

CORRECT RESULT

$60,000

06 Inventory Turnover Ratio

Question data

- Cost of Goods Sold = $150,000
- Average Inventory = $25,000

Supplied solution

25,000 / 150,000 = 0.16 times

✕ MISTAKE IDENTIFIED

The formula was reversed. COGS should be divided by average inventory, not the other way around.

✓ CORRECT FORMULA

Inventory Turnover = Cost of Goods Sold ÷ Average Inventory

Reasoning

The ratio measures how many times average inventory is turned into cost of goods sold during the period.

✓ CORRECT RESULT

6 times

07 Debt to Equity Ratio

Question data

- Total Debt = $200,000
- Shareholders' Equity = $100,000

Supplied solution

100,000 / 200,000 = 0.5:1

× **MISTAKE IDENTIFIED**

The formula was reversed. Debt is the numerator and shareholders' equity is the denominator.

✓ **CORRECT FORMULA**

Debt-to-Equity Ratio = Total Debt ÷ Shareholders' Equity

Reasoning

This compares financing supplied by creditors with financing supplied by owners.

✓ **CORRECT RESULT**

2:1

08 ROCE

Question data

- Operating Profit = $60,000
- Capital Employed = $300,000

Supplied solution

300,000 / 60,000 × 100 = 500%

✗ MISTAKE IDENTIFIED

The numerator and denominator were reversed. Operating profit should be divided by capital employed.

✓ CORRECT FORMULA

ROCE = (Operating Profit ÷ Capital Employed) × 100

Reasoning

ROCE measures the return generated from the capital employed, so the return goes over the capital base.

✓ CORRECT RESULT

20%

09 EPS

Question data

- Net Profit = $100,000
- Number of Shares = 25,000

Supplied solution

25,000 / 100,000 = 0.25; Answer: $0.25 per share

✕ **MISTAKE IDENTIFIED**

The formula was reversed. Earnings per share equals profit divided by the number of shares.

✓ **CORRECT FORMULA**

EPS = Net Profit ÷ Number of Shares

Reasoning

Divide total earnings among the number of shares to find earnings attributable to each share.

✓ **CORRECT RESULT**

$4.00 per share

10 Accounts Receivable Turnover Ratio

Question data

- Net Credit Sales = $400,000
- Average Accounts Receivable = $80,000

Supplied solution

80,000 / 400,000 = 0.2 times

× **MISTAKE IDENTIFIED**

The formula was reversed. Net credit sales should be divided by average accounts receivable.

✓ **CORRECT FORMULA**

Receivables Turnover = Net Credit Sales ÷ Average Accounts Receivable

Reasoning

The ratio indicates how many times receivables are converted through credit sales during the period.

✓ **CORRECT RESULT**

5 times

Final Correction Summary

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Key lesson: Most errors in the supplied document come from either reversing a ratio formula or applying the wrong operation. A reliable approach is to write the formula first, identify what belongs in the numerator and denominator, then perform the arithmetic.